

Bayesian methods for the evaluation and limitation of disclosure risk in official statistics

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UNECE EXPERT MEETING ON STATISTICAL DATA CONFIDENTIALITY

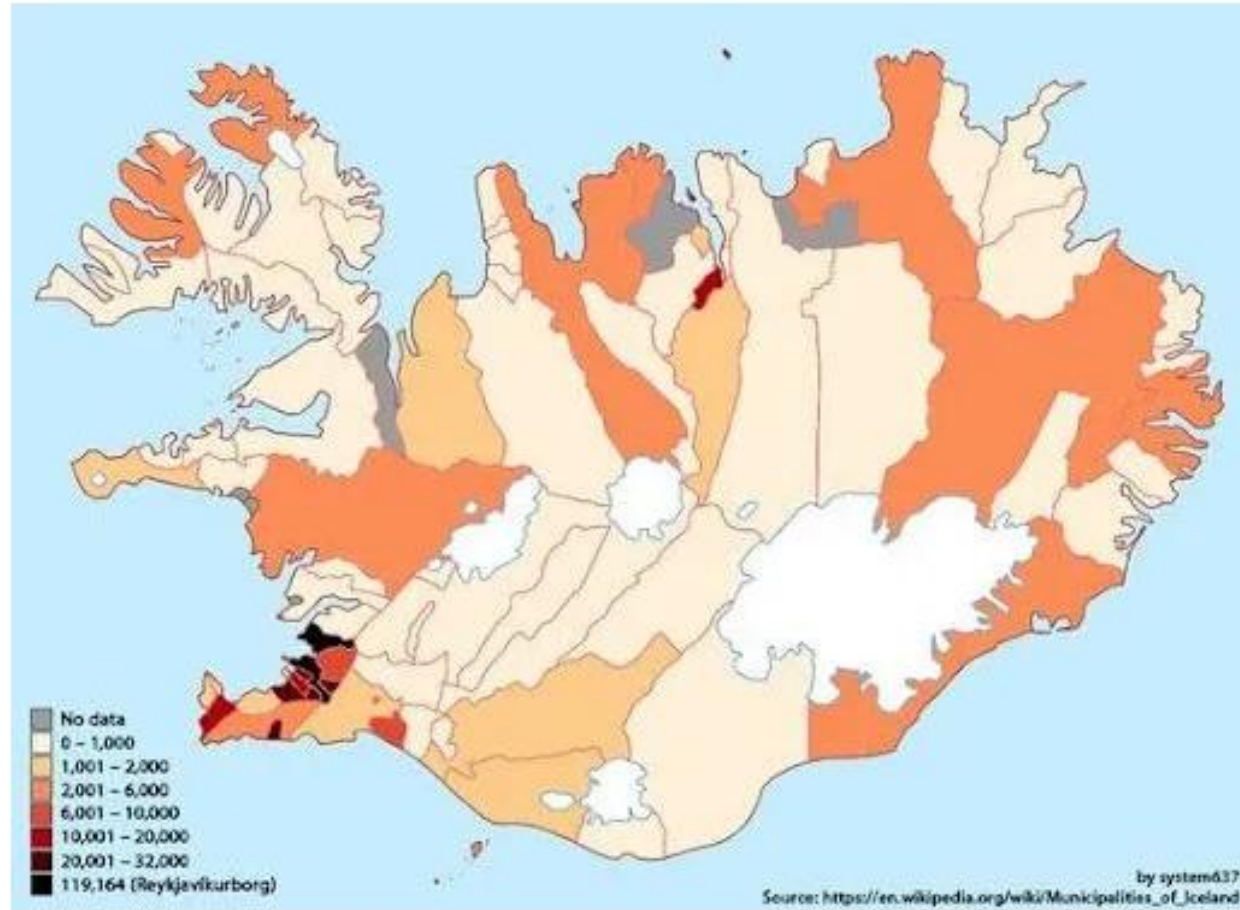
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Auditorium of the Poblenou Campus of Universitat Pompeu Fabra, Roc Boronat 138, 08018 Barcelona

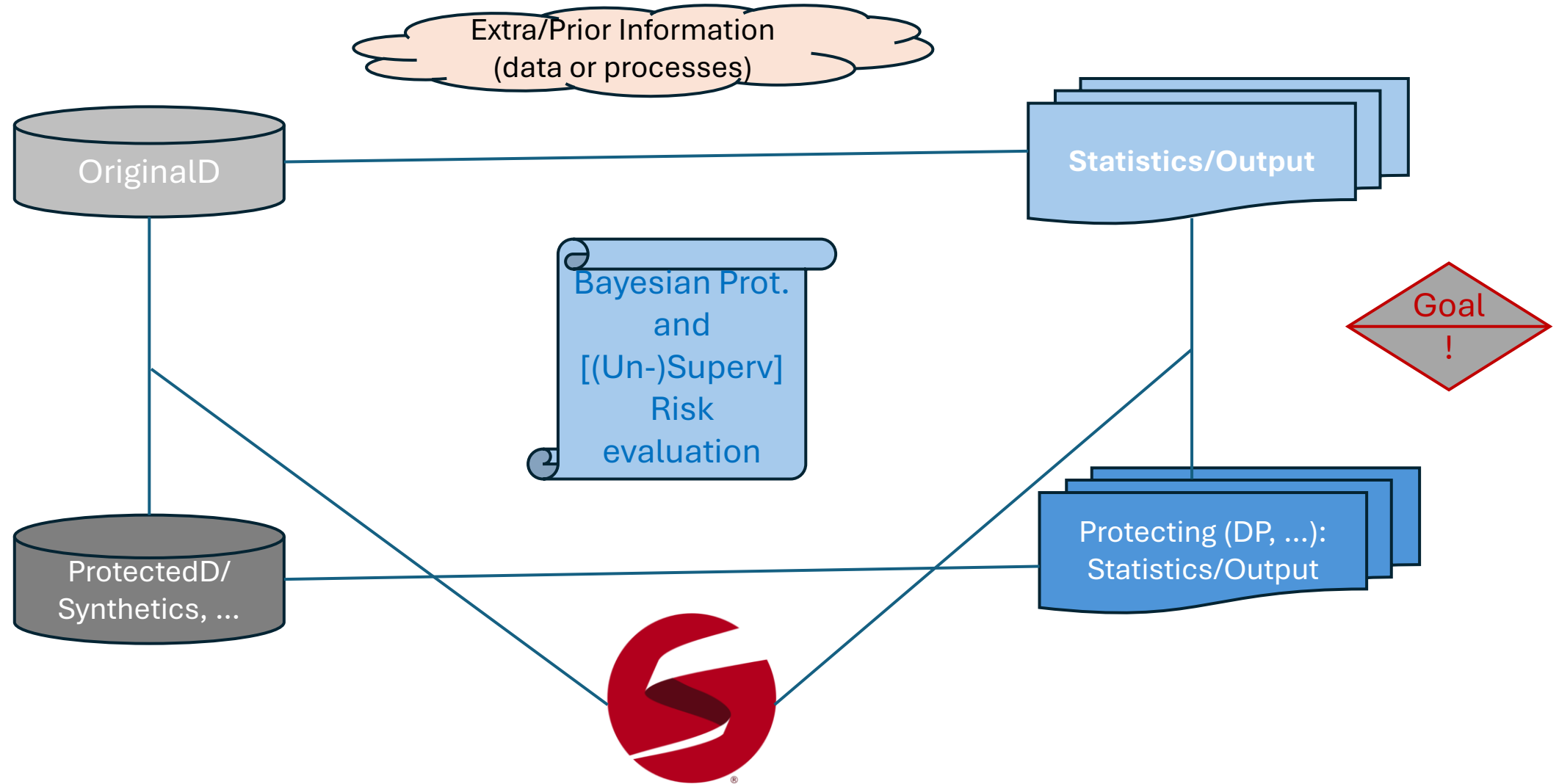
Official Statistics: specific features of SDC

- Demands of internal experts and external researchers
- Requirements of SDC experts
- Novel theoretical and experimental developments - *evaluation*

Icelandic motivation



Bayesian solutions



Consequence: options

- Bayes **risk** for: Original /Synthetic / protected Data
- Bayes building: synthetic data & [Differentially] Private output
- ML/BDeepL: for synthesizing/protecting & risk evaluation
- BDeepL+DP
- Anomaly for rare patterns / unsupervised risk detection
- Validation connection*

Status

- generating multiple synthetic data sets (Graham and Penny, 2005; Graham et al., 2009; Drechsler and Reiter, 2010, 2012)
- simultaneous detection and correction of errors as well as disclosure limitation for microdata (Kim, Reiter and Karr, 2016)
- differentially private algorithms through posterior sampling (Dimitrakakis et al, 2017)
- analytical/numerical calculation of DP parameters for given privacy budget (*Kazan and Reiter, 2024*)

New focus and new solutions

- the *risk* quantification and the solutions offered by Bayesian formulations for *nuanced* assessment of attacks and risks
- *non-parametric* hierarchical Bayesian models for building data protection procedures and evaluating the disclosure risk
- Bayesian *DP*
- *unsupervised* anomaly detection as tool for disclosure risk detection/DVal

Evaluation process:

1. Bayesian risk measures and optimum sdc/DP - parameters
2. Tools
3. [B, DeepN] synthetics
4. Unsupervised disclosure risk detection

1. Bayes and the (DP) privacy budget: risk measures, prior and posterior prob.

Under adversary model & extra-knowledge, for an arbitrary unit:

$$prior_{id} = P(discl_{id})$$

$$prior_{attr} = P(discl_{attr})$$

Relative disclosure risk as posterior/prior ratio --->

optimum curve $risk = F(discl_{id}, discl_{attr})(unit)$, for all units from bounds of (w.r.t. the risk-profile):

$$\frac{P(discl_{id}, discl_{attr} | output_{released})}{P(discl_{id}, discl_{attr})} =$$

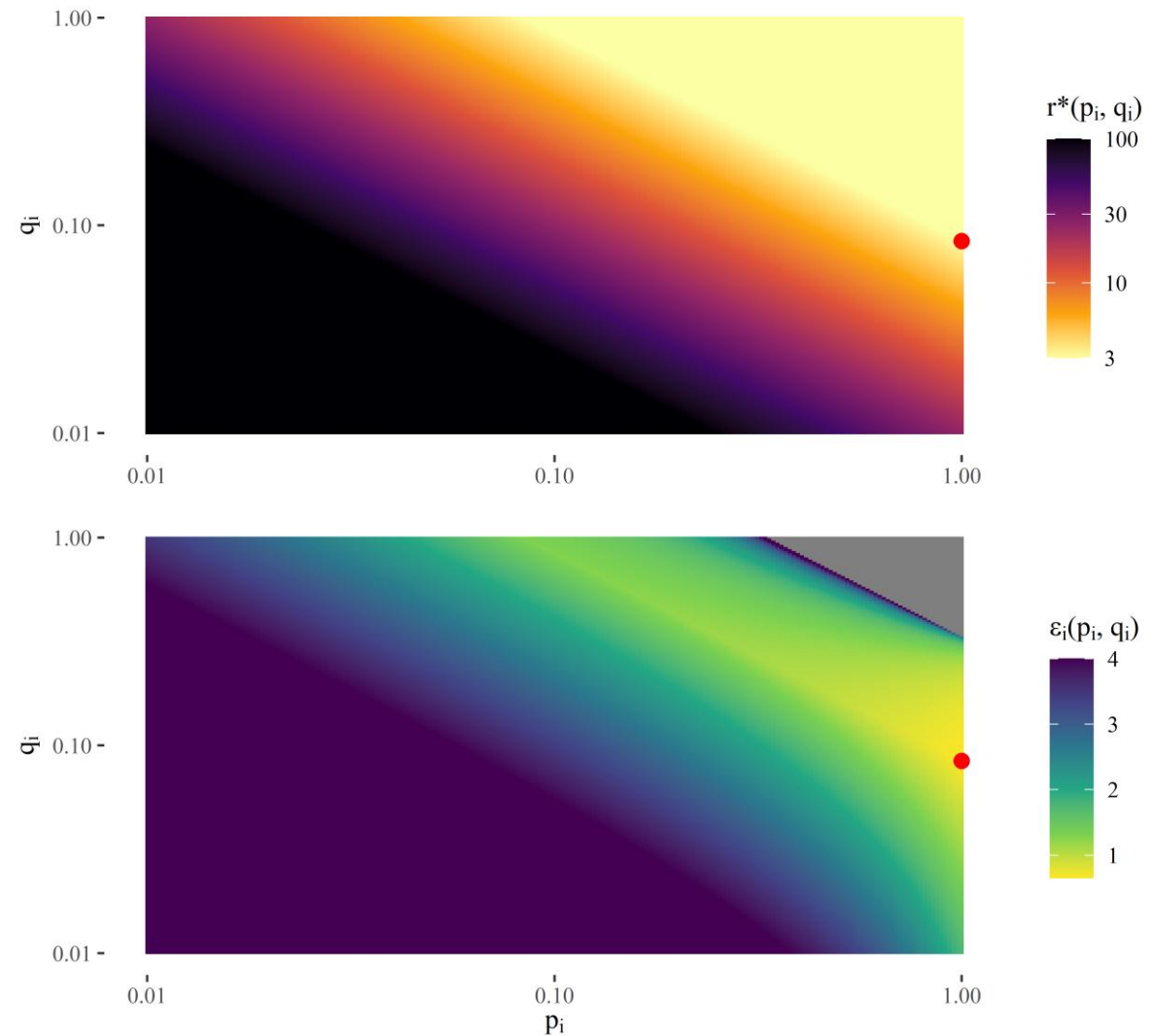
$$\frac{P(discl_{attr} | discl_{id}, output_{released})}{P(discl_{attr} | discl_{id})} \cdot \frac{P(discl_{id} | output_{released})}{P(discl_{id})}$$

A **solution** example for protection-parameter choice:

„Prior-itzing Privacy: A Bayesian Approach to Setting the Privacy Budget in Differential Privacy”
(Kazan and Reiter, 2024)

p -identification; q -attribute disclosure

• - optimum



2. Tools:

<https://github.com/sdcTools>

& also:



- **R** -Synthetics:

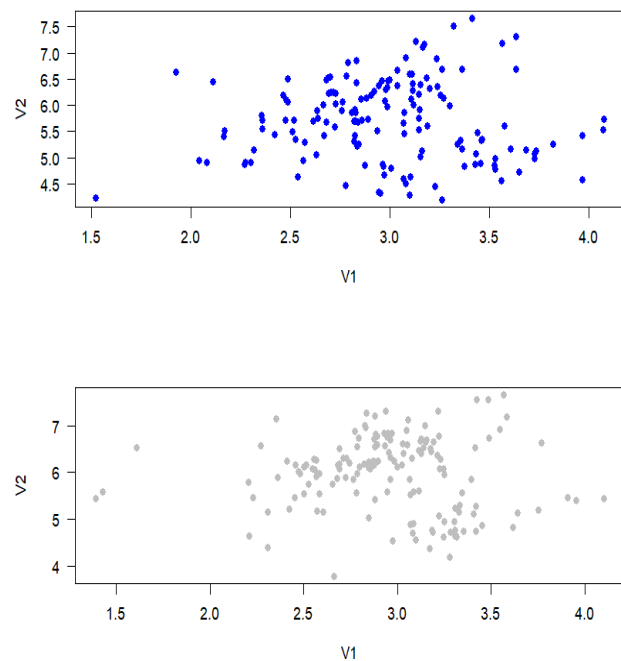
NPBayesImputeCat, synMicrodata, synthpop, RGAN, synthesizer

- **R**-Differential Privacy: *Dppack, diffpriv*

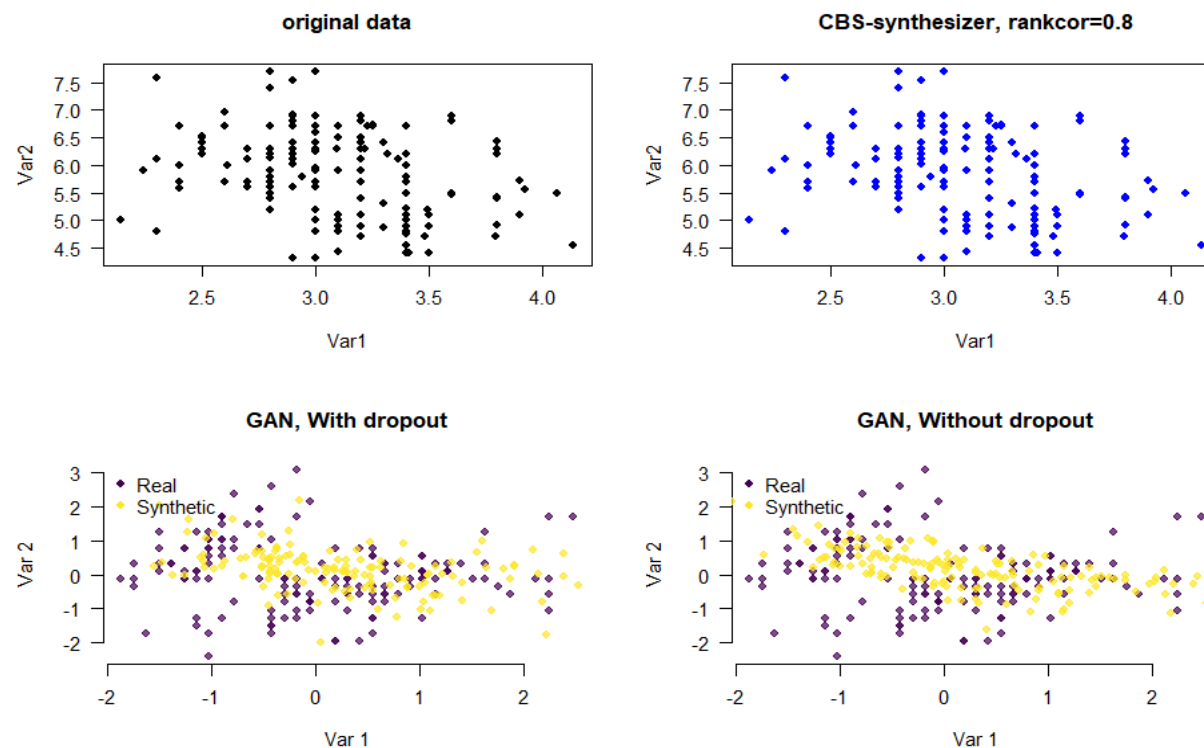
- *Risk, sdc-parameters & <DL with DP>*

3. Synthetics

Bayesian synthetics

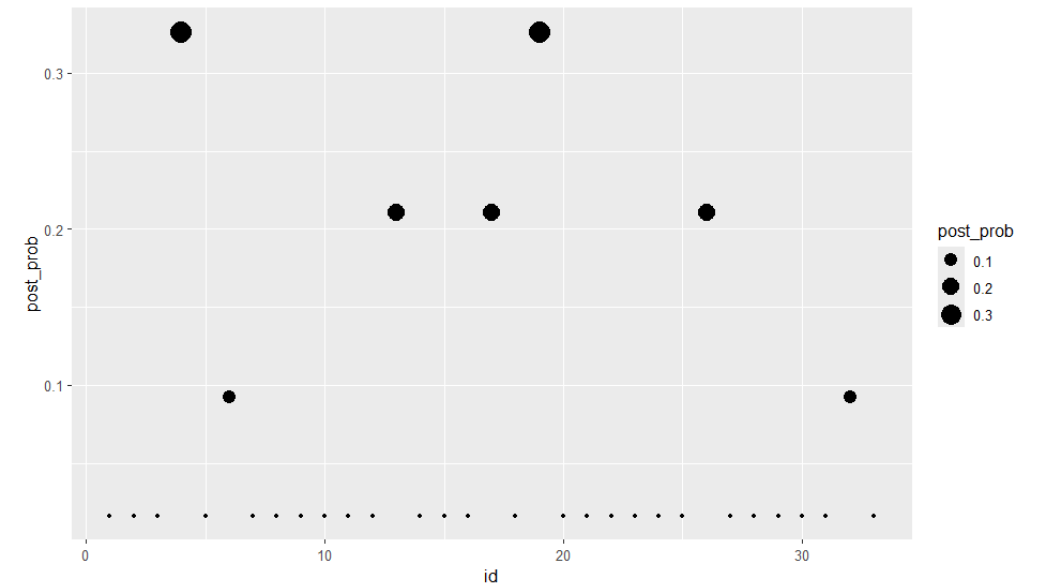
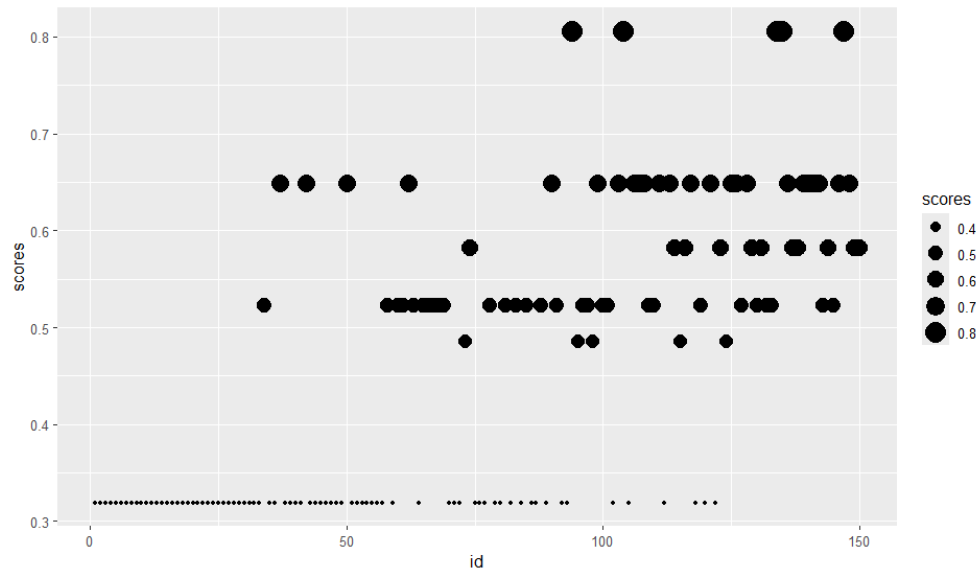
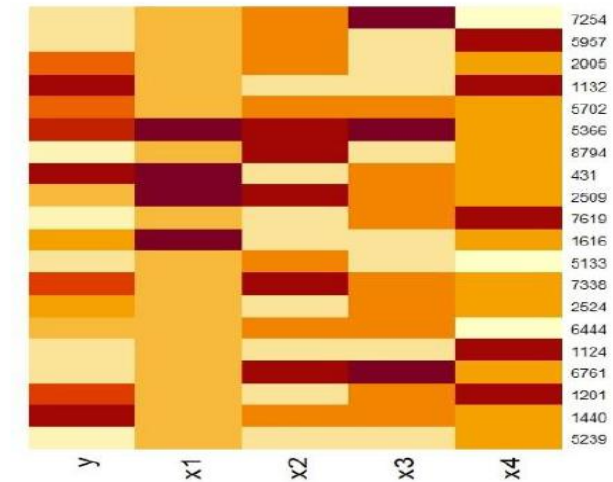


Newer synthetics



4. *Unsupervised risk detection:*

*Rare patterns and anomalies,
scores and posterior probabilities*



Note

- Note to the UNECE Secretariat:

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Thank you !

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- <https://github.com/violetacln/testingSDCtools>